Hafnium-Based Ferroelectric Diode with Interlayer Enhancement for In-Memory Logic Application

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***Abstract*—Logic-in-Memory computing (LiM) has emerged as a key approach to overcome the von Neumann bottleneck, and hafnium-based ferroelectric diode (Fe diode) is a promising candidate for this application. For further performance optimization, we fabricated Fe diode devices with two strcutures: TiN/HZO/TiN and interlayer-enhanced TiN/HZO/HfO_2/TiN. The HfO_2 interlayer devices exhibit improved retention characteristics while maintaining bidirectional rectification behavior and a well-defined logic window, enabling stable logic operations. We demonstrated a circuit scheme of implementing both IMP and NIMP logic functions in a single operation with an ultra-low operating power consumption of 20.65 aJ. Additionally, by configuring different device combinations, all 16 Boolean logic functions can be realized based on Fe diodes. This work highlights the potential of interlayer-enhanced Fe diodes, offering a simple, easy-to-integrate, and low power consumption approach for developing high-performance LiM systems.**

Keywords—Ferroelectric diodes; interlayer engineering; logic-in-memory; bidirectional rectification

I. INTRODUCTION

The rapid development of artificial intelligence and big data has driven exponential increase in data processing demands, exposing the inherent limitations of traditional Von Neumann computing architecture, specifically the "Von Neumann bottleneck" caused by the separation of memory and processing units [1, 2]. Logic-in-Memory (LiM) computing has emerged as a promising paradigm by integrating data memory and logic operations within the same device thereby significantly reducing power consumption and latency associated with data movement [3-5].

Among various device candidates, ferroelectric diode (Fe diode) emerges as a promising solution for next-generation low power consumption LiM computing systems. This two-terminal device boasts a simple structure and self-rectification characteristics, as well as the capacity to support high-density crossbar array memory technology without the need for selectors [4, 6]. Despite the extensive utilization of various ferroelectric materials in electronic devices, the poor compatibility of conventional ferroelectric materials with CMOS processes and the difficulty of scaling down the size pose challenges for their application in advanced computing architectures. Conventional ferroelectric materials can only maintain stable remanent polarization (P_r) and low leakage current at thicknesses of several dozen nanometers or greater. Reducing the thickness leads to unstable polarization states, and the film thickness limits the scalability of conventional ferroelectric devices. Additionally, conventional ferroelectric materials require high fabrication and annealing temperatures and contain polluting elements such as Pb and Bi, making them incompatible with CMOS processes [3]. In contrast, hafnium-based ferroelectric materials discovered for the first time in 2011 that exhibit ferroelectricity at room temperature hold promise for addressing these challenges [7].

Hafnium-based ferroelectric materials have been demonstrated to maintain a stable polarization state at a thickness as low as 10 nm. Furthermore, hafnium oxide, as a high-dielectric-constant material, has been extensively utilized in processor applications and is already extensively employed in Si-based CMOS devices, with mature material properties and fabrication processes [8-10]. Additionally, the intrinsic polarization characteristics of hafnium-based ferroelectric materials are capable of enhancing switching speed and

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reducing power consumption of logic operation, aligning with the core requirements of LiM systems [11, 12]. However, hafnium-based Fe diodes still have potential for performance enhancement in LiM applications, such as the retention characteristics and memory window stability. Interface engineering can be an efficient approach by introducing functional transition layers at the electrode/ferroelectric interface to control the interface band and suppress charge injection [13-16]. Moreover, extant research on Fe diodes-based logic circuit primarily focuses on utilizing voltage signals as the main input for logic operations [13, 17]. Due to the lack of internal storage capabilities to preserve input states, external memory units or signal conversion circuits must be integrated to enable logic functions, increasing system complexity and energy consumption.

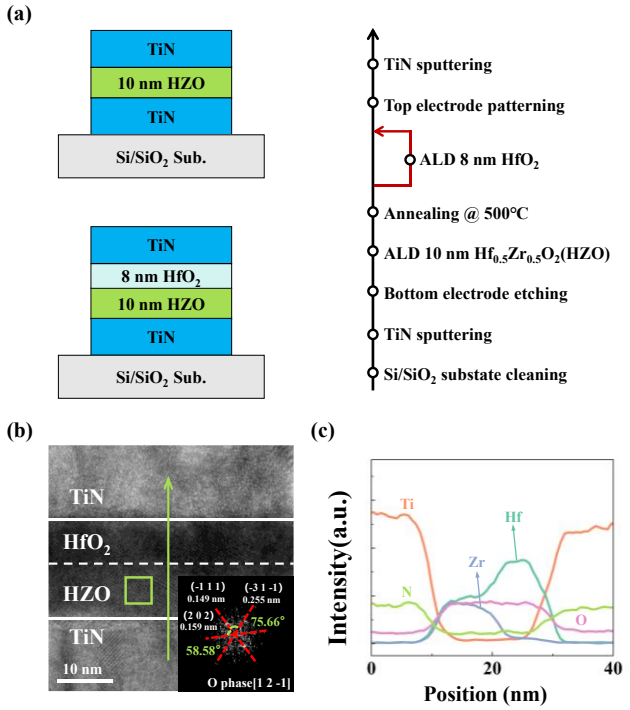


Fig. 1. (a) Schematic diagrams and fabrication process flow of the Fe diode devices. (b) High-resolution transmission electron microscopy (HRTEM) image of the HfO_2 interlayer device, with the inset showing the Fast Fourier transform (FFT) pattern. (c) Energy-dispersive X-ray spectroscopy (EDS) line-scan image of the HfO_2 interlayer device.

In this work, we proposed hafnium-based Fe diodes with interlayer enhancement for LiM applications. Fe diode devices with two structures were fabricated, the TiN/HZO/TiN structure as a control sample and an interlayer-enhanced TiN/HZO/HfO_2/TiN structure. An investigation was conducted into the effect of the HfO_2 interlayer on device performance, leading to the identification of enhancements in characteristics such as retention and memory window performance. Then we demonstrated the Fe diode-based logic circuit scheme where both input and output variables are directly represented by resistance states, avoiding additional signal conversion. We implemented both IMP (implication) and NIMP (negative implication) logic functions in a single circuit with an ultra-low operating energy of 20.65 aJ. Additionally, by configuring different device combinations, all 16 Boolean logic functions can be realized. This work highlights the potential of Fe diode with interlayer enhancement for high-performance, low-power and low-complexity LiM systems.

II. Experiments

Fig. 1(a) schematically illustrates the structure and fabrication process of the Fe diode devices proposed in this work. The devices consist of two structures, one of which is TiN/HZO/TiN as a control sample, and the other is TiN/HZO/HfO_2/TiN. The Si/SiO_2 substrate was sequentially cleaned with acetone, ethanol and deionized water. The TiN bottom electrode (BE) was then deposited via magnetron sputtering, followed by patterning through UV lithography and etching. A 10 nm thick HZO film was deposited at 280 °C by atomic layer deposition (ALD), where the cycle ratio of HfO_2:ZrO_2 was maintained at 1:1. TEMAH, TEMAZ, and H_2O were used as precursors for Hf, Zr, and O, respectively. The samples were subjected to rapid thermal annealing (RTA) at 500 °C for 30s in a N_2 atmosphere to induce the formation of a ferroelectric phase. Subsequently, an 8 nm thick HfO_2 layer was deposited by ALD. Then the TiN were deposited by magnetron sputtering and patterned as top electrode (TE).

III. Results and Discussion

The high-resolution transmission electron microscopy (HRTEM) image of the cross-section of the HfO_2 interlayer device is shown in Fig. 1(b), revealing clear layer interfaces and distinct crystal lattice of the HZO film. The Fast Fourier transform (FFT) pattern of the green boxed region is shown in the inset, and analysis confirms the orthorhombic (o) phase of the HZO layer, indicating ferroelectric properties [4, 8, 18]. Additionally, we analyzed the elemental distribution across layers of the device along the green scan path in Fig. 1(b) using energy-dispersive X-ray spectroscopy (EDS). This analysis verified the device's basic structure and confirmed the interface thickness observed in Fig. 1(a).

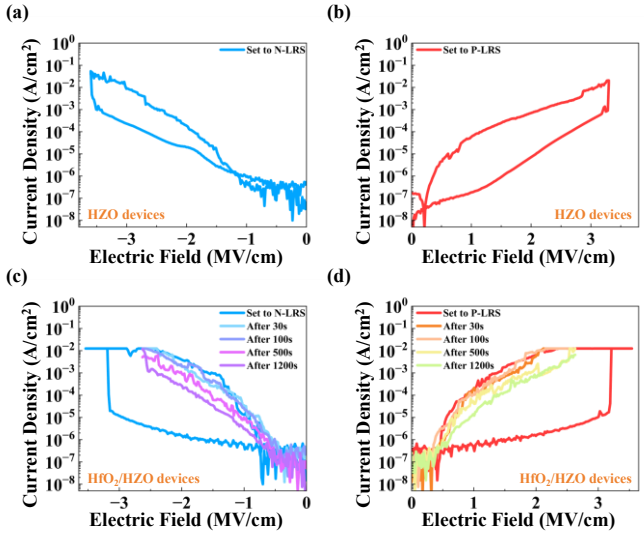


Fig. 2. Current density-electric field (J-E) curves of single HZO device under (a) negative bias and (b) positive bias. Current density-electric field (J-E) curves of HfO_2 interlayer device under (c)negative bias and (d) positive bias.

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Fig. 2(a)–(d) illustrate the current density–electric field (J–E) curves obtained by direct current (DC) sweep for Fe diode devices with single HZO and HfO_2/HZO stack. In Fig. 2(a) and (b), under the applied electric field, the single HZO device transitions from high-resistance state (HRS) to low-resistance state (LRS), exhibiting a memory window which arises from bidirectional rectification behavior. The red curve represents the LRS under positive bias (P-LRS), while the blue curve corresponds to the N-LRS under negative bias. Fig. 2(c) and (d) show the scanning results for the HfO_2 interlayer device. Conversely, the device with the interlayer exhibits a distinct memory window. Subsequently, to assess the retention characteristics of the HfO_2 interlayer device, after setting devices to the LRS, we conducted additional scans at varying time intervals following their configuration to the LRS. The scan curves after 30s, 100s, and 500s in Fig. 2(c) and (d) closely resembled the curves in the LRS, while the curve after 1200s exhibits a slight decline, indicating the reliable retention characteristic of the interlayer device for LiM applications. Consequently, it can be substantiated that the HfO_2 interlayer enhances device performance [13]. Then the electrical characterization was conducted on the TiN/HZO/HfO_2/TiN device. The polarization-electric field (P–E) loop shown in Fig. 3(a) was obtained via positive-up negative-down (PUND) measurements, which confirms the ferroelectric behavior of the device. The relatively low P_r values observed are attributed to the absence of a capping metal during HZO annealing. Without the compressive stress imposed by a capping layer, the HZO film likely exhibits a lower fraction of ferroelectric orthorhombic phase, thereby lowering the P_r. Despite this, omitting the capping layer simplifies the fabrication process by eliminating the need for capping layer deposition and removal steps, while still enabling sufficient crystallization and ferroelectricity in the HZO film.

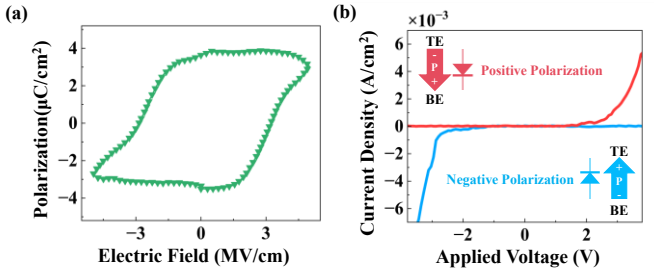


Fig. 3. (a) Polarization-electric field (P–E) loop of the HfO_2 interlayer device. (b) Bidirectional rectification characteristics of the HfO_2 interlayer device.

As previously mentioned, Fe diode devices exhibit two low-resistance states: P-LRS and N-LRS. Therefore, after the device transitions from the high-resistance state under positive bias (P-HRS) to P-LRS, a positive scanning voltage was applied. The red curve in Fig. 3(b) illustrates the diode characteristics with unidirectional rectification of the device under positive polarization. When it was set to N-LRS, the rectification direction switched under negative scanning voltage. The corresponding polarization direction and diode rectification direction are illustrated by the blue curve in Fig. 3(b). The device exhibited switchable diode characteristics, indicating that the current transmission direction is programmable. Notably, the device's coercive field (E_c) is comparable to the electric field of resistive state transitions. This overlap suggests that resistive switching in the device may be regulated by the ferroelectric polarization of HZO. The polarization charge generated by the polarization of the ferroelectric film affects the interface barrier, thereby causing changes in the energy band and resulting in different resistive states.

From the perspective of energy bands, when the HZO layer undergoes polarization, its energy bands bend according to the direction of the polarization. With no applied voltage, the conduction band energy level on the polarization direction side remains elevated, forming an electronic potential barrier that hinders electron injection. Meanwhile, the positively polarized charges on the opposite polarization direction side attract electrons, causing them to spontaneously accumulate at the interface. When a voltage is applied in the same direction as polarization, the height of this barrier is reduced, allowing electrons to rapidly inject into the region opposite the polarization direction, thereby generating a significant current and exhibiting LRS. Conversely, when the voltage is applied in the opposite direction to the polarization, the amorphous structure of the HfO_2 layer increases the barrier height, suppressing leakage current and enhancing barrier isolation. The elevated barrier significantly impedes electron injection, resulting in extremely low conductive current to maintain the stability of the memory window and retention characteristics of the device [13].

Based on the bidirectional rectification characteristics of Fe diode devices, we proposed circuit schemes for Boolean logic computation. As shown in Fig. 4(a), the circuit structure can realize IMP and NIMP logic operations in one step. In this scheme, the resistive states (LRS/HRS) of devices A and B serve as the two logic input variables. The TEs of device A and device B are connected to the same word line (WL), and their BEs are connected to different bit lines (BL). By applying excitation voltage V to the BL of device A and ground to the BL of device B, a series circuit was formed. Fig. 4(b) illustrates the applied voltage scheme in the crossbar array, and the BLs of non-selected devices were biased with V/2 to remain in their initial state without interfering with the target series circuit.

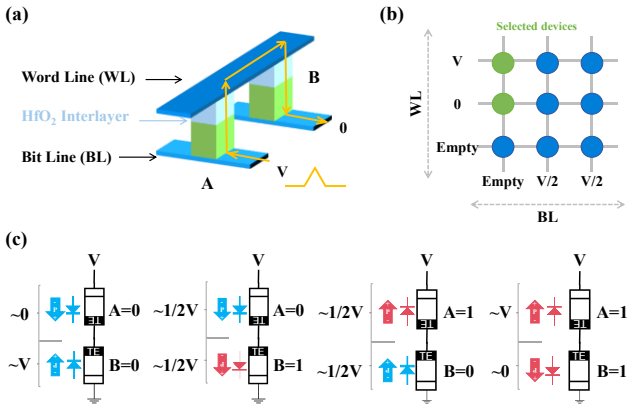


Fig. 4. (a) Schematic diagram of the logic circuit scheme based on Fe diode. (b) The scheme of applied voltage in the crossbar array. (c) Equivalent circuit diagrams of the logic circuit under different inputs.

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Fig. 4(c) shows the equivalent circuit diagram of the series circuit under different input combinations. We define the direction from the TE to the BE as the direction of logic value read. When the device is under positive polarization, it exhibits LRS along the read direction, representing logic 1. When the device is under negative polarization, it exhibits HRS along the read direction, representing logic 0. Therefore, when input A = 0 and B = 0, the equivalent diode of device A conducts in the direction of the excitation voltage V, exhibiting LRS in the logic write circuit. However, the conduction direction of device B is opposite to the excitation voltage, exhibiting HRS in the circuit. Therefore, the voltage of the two devices in the circuit is ~0 and ~V, respectively. When input A = 0 and B = 1, both device A and device B exhibit LRS in the circuit, with equal voltage division, 1/2 V. Similarly, when input A = 1 and B = 0, both devices exhibit HRS in the circuit, with the same voltage of 1/2 V. When A = 1 and B = 1, the conduction direction of device A is opposite to the excitation voltage, exhibiting HRS, while device B exhibits LRS, resulting in voltage divisions of ~V and ~0, respectively.

Subsequently, the logic circuit scheme was validated under different input conditions through simulations. After applying excitation voltage to the logic write circuit, the voltage-current curves of the input/output states and the corresponding current responses of circuit are illustrated in Fig. 5.

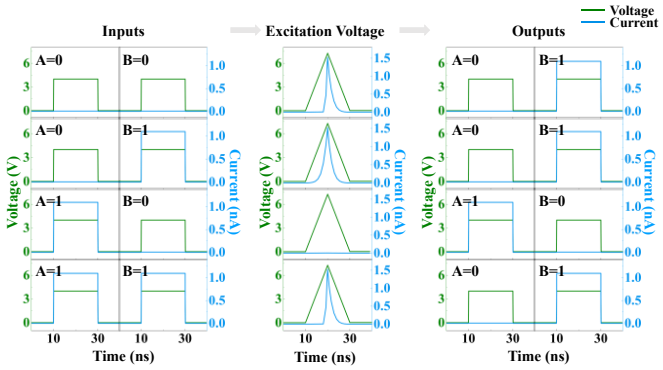


Fig. 5. Voltage-current curves of the logic write circuit.

When inputs A and B were both set to 0, two devices exhibited LRS and HRS respectively. In the series circuit, the voltage of device B was ~V, which exceeded the resistance state transition threshold. Consequently, the state of B switched, and the circuit exhibited a steep current peak, as shown in the first line of Fig. 5. Similarly, if A = 0 and B = 1, both devices maintained the LRS. The voltage of device A and device B was ~ 1 / 2V. Since the voltage was below the transition threshold, the input logic value did not change, and the current response of the circuit was a steadily rising current peak. Conversely, when A = 1 and B = 0, the two devices were in HRS in the circuit, resulting in an equivalent voltage. The current response was minimal. Finally, when both devices were configured to 1, the voltage of device A was ~V, resulting in the state change and the steep current peak. The results of these simulations consistently validate the logic operation mechanism proposed in the circuit design. A comprehensive analysis of input-output states confirms that this circuit design enables implementation of IMP and NIMP logic in a single operation, with results stored in devices B and A, respectively. After simulation, the average power consumption of four input conditions was as low as 20.65 aJ. Furthermore, all sixteen Boolean logic functions can be realized using Fe diodes-based circuit. As shown in Table I, in comparison to other work, the Fe diode-based logic cell proposed in this work features a simple structure without additional devices to be cascaded. In addition, power consumption of the logic scheme is as low as the aJ level. The input data is stored as resistance values within the device, enabling non-volatile storage without requiring additional form conversion. Fe diode devices in this work inherently offer faster response speed and ultra-low power consumption, providing core technological support for next-generation low-power LiM applications.

TABLE I. PERFORMANCE COMPARISON OF VARIOUS POTENTIAL LIM DEVICES

Logic cell structure	TiN/HZO/TiN/ HfO_2/IGZO	Ti/$HfSe_xO_y$/ $HfSe_2$/Au	TiN/HfO_x/Ti/ TiN/Ti/HfO_x/ TiN	TiN/HZO/ HfO_2/TiN
Boolean Logic Functions Realized	14	3 (XOR, IMP, NAND)	16	16
Power Consumption	aJ level	fJ level	pJ level	aJ level
Input Data Storage in Device	No (input in the form of voltage)	Yes	Hybrid (in the form of voltage and resistance)	Yes
Additional Device Required	Yes (with n-type load transistor)	No	Yes (with load resistance)	No
Reference	[17]	[19]	[20]	This work

IV. CONCLUSION

In summary, this work has improved the performance of hafnium-based Fe diodes in LiM applications through interface engineering and proposed logic circuit schemes. By comparing Fe diode devices with TiN/HZO/TiN structure and TiN/HZO/HfO_2/TiN structure, enhancement in data retention characteristics and memory window stability were observed in the HfO_2 interlayer devices, which is key advancement for achieving stable LiM operations. Based on the device with interlayer enhancement, we proposed a logic circuit scheme, facilitating parallel computation of IMP and NIMP logic functions with a power consumption of ~20.65aJ. Furthermore, different circuit combinations can achieve scalabilities for all 16 Boolean operations. These results validate interface engineering as a viable strategy for enhancing the performance of hafnium-based Fe diodes, providing insights for next-generation low-power LiM computing architectures.

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